

Capstone

SpaceX Falcon 9 First Stage

Landing Prediction

Duration: 2026.06.02- 2026.06.09

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Resource provided by IBM Skills Network
through IBM Data Science Professional Certification

Executive Summary

- According to the historical recode of SpaceX Falcon 9 Landing plan, we could design and validate models to predict Falcon 9 landing probability through collected data and methodology.
- First of all, I collected data to identified variables and selected necessary attributes for modeling preparation using Python libraries .
 - Using correlation method to validate the numerical variables .
 - Plotting their relationship through scatter and checking whether they are linear or non-linear
 - Check the relationship between pay load mass (kg) and Launch sites
 - Compared three variables (Success rate, Flight Number, and Pay Load Mass) with orbit types
 - Plotting the yearly trend of the launch success
- Secondly, I sued Folium to identify the launch site locations in the map with markers to show the success indicators
- Finally, I built four machine learning models to validate the best hyperplane in model prediction through accuracy scores and confusion matrix. There are the machine learning models I chose to validate:
 - Logistic Regression
 - SVM
 - Decision Tree
 - KNN

Agenda

➤ **Phase 1 – Exploring and Preparing Data**

- Data wrangling
- Exploratory Data Analysis
- Exploratory Data Analysis with Visualization- Launch Site in the Map with Location Markers

➤ **Phase 2 – Machine Learning Prediction**

- Table of Results with four machine learning methods tested
- Conclusion

Phase 1 – Exploring and Preparing Data

Selected variables

- Checking variables and their types
 - Libraries

```
import numpy as np
import pandas as pd
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

- df.info()

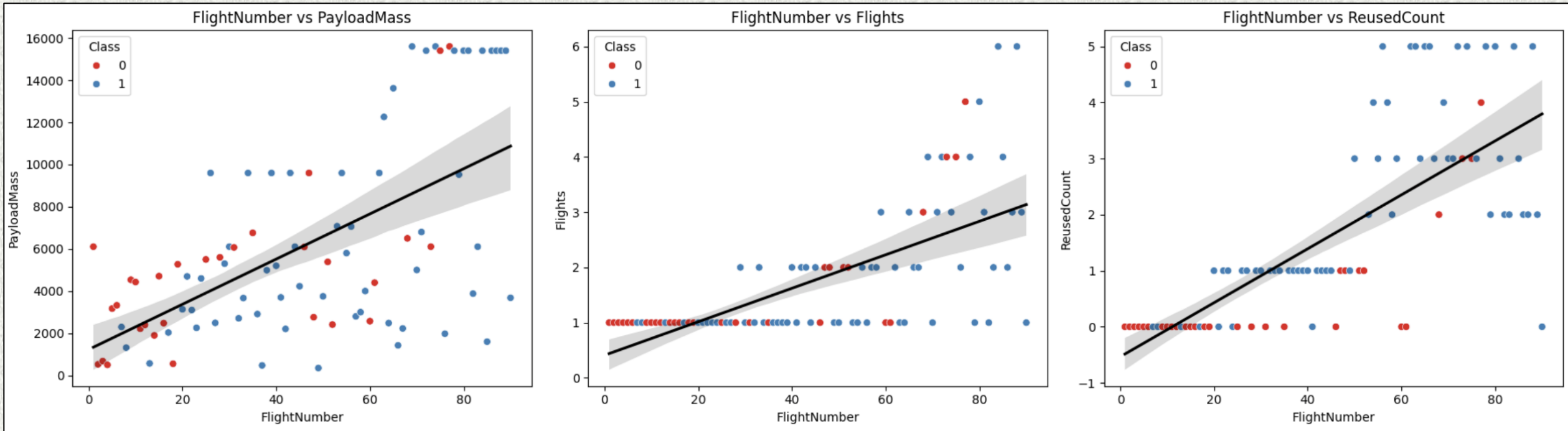
```
<class 'pandas.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   FlightNumber          90 non-null    int64
1   Date                  90 non-null    str
2   BoosterVersion        90 non-null    str
3   PayloadMass           90 non-null    float64
4   Orbit                 90 non-null    str
5   LaunchSite            90 non-null    str
6   Outcome               90 non-null    str
7   Flights               90 non-null    int64
8   GridFins              90 non-null    bool
9   Reused                90 non-null    bool
10  Legs                  90 non-null    bool
11  LandingPad            64 non-null    str
12  Block                 90 non-null    float64
13  ReusedCount           90 non-null    int64
14  Serial                90 non-null    str
15  Longitude             90 non-null    float64
16  Latitude              90 non-null    float64
17  Class                 90 non-null    int64
dtypes: bool(3), float64(4), int64(4), str(7)
memory usage: 10.9 KB
```

Using
corr()
function to
check their
correlation

The result of Corr ()

	FlightNumber	PayloadMass	Flights	ReusedCount	Class
FlightNumber	1.000000	0.596307	0.652842	0.735193	0.401923
PayloadMass	0.596307	1.000000	0.665218	0.612327	0.199992
Flights	0.652842	0.665218	1.000000	0.684802	0.149787
ReusedCount	0.735193	0.612327	0.684802	1.000000	0.466584
Class	0.401923	0.199992	0.149787	0.466584	1.000000

Plot their Relationships

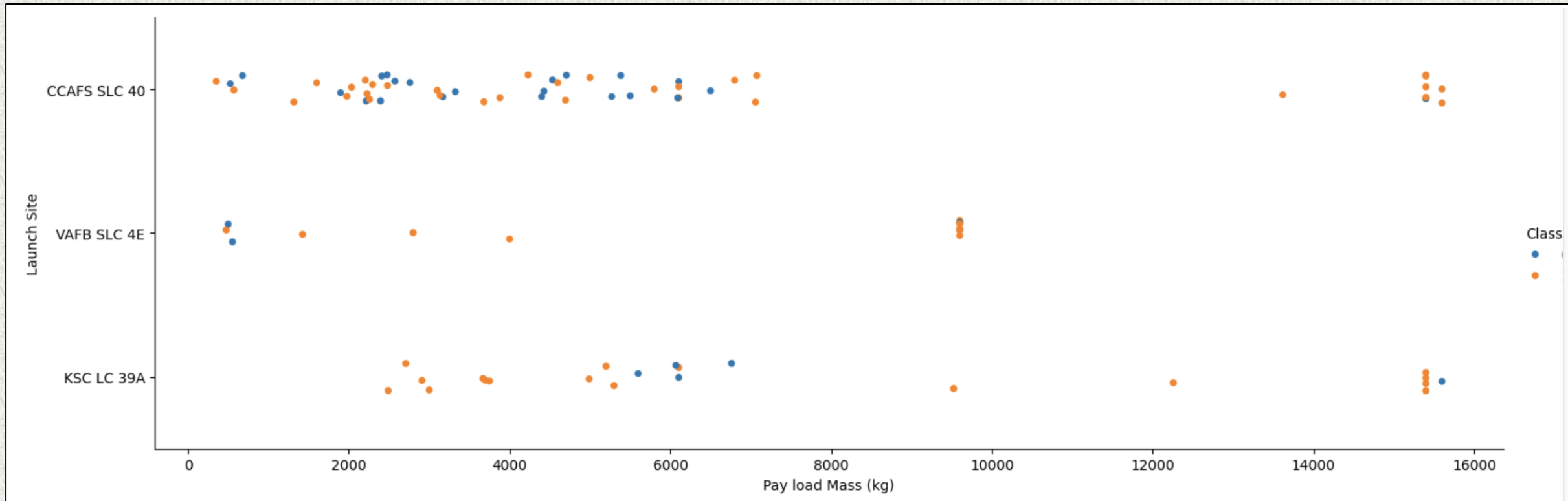


Class: 1 – Success
Class: 0 -- Failed

- According to the figures of Flight Number & Flights, and Flight Number & Reused Count, they shows no linear relationships. As result, I decided to remove these two combination variables.
- Another correlation between Flight Number & Pay Load Mass I would validate them with EDA and other plots to ensure their correlations.

Selected variables: Pay Load Mass (Kg) & Launch Site

```
sns.catplot (y='LaunchSite', x= 'PayloadMass', hue='Class', data=df, aspect=3)  
plt.xlabel ("Pay load Mass (kg)", fontsize= 10)  
plt.ylabel ("Launch Site", fontsize= 10)  
plt.title ("The relationship between Pay load Mass ")  
plt.show()
```



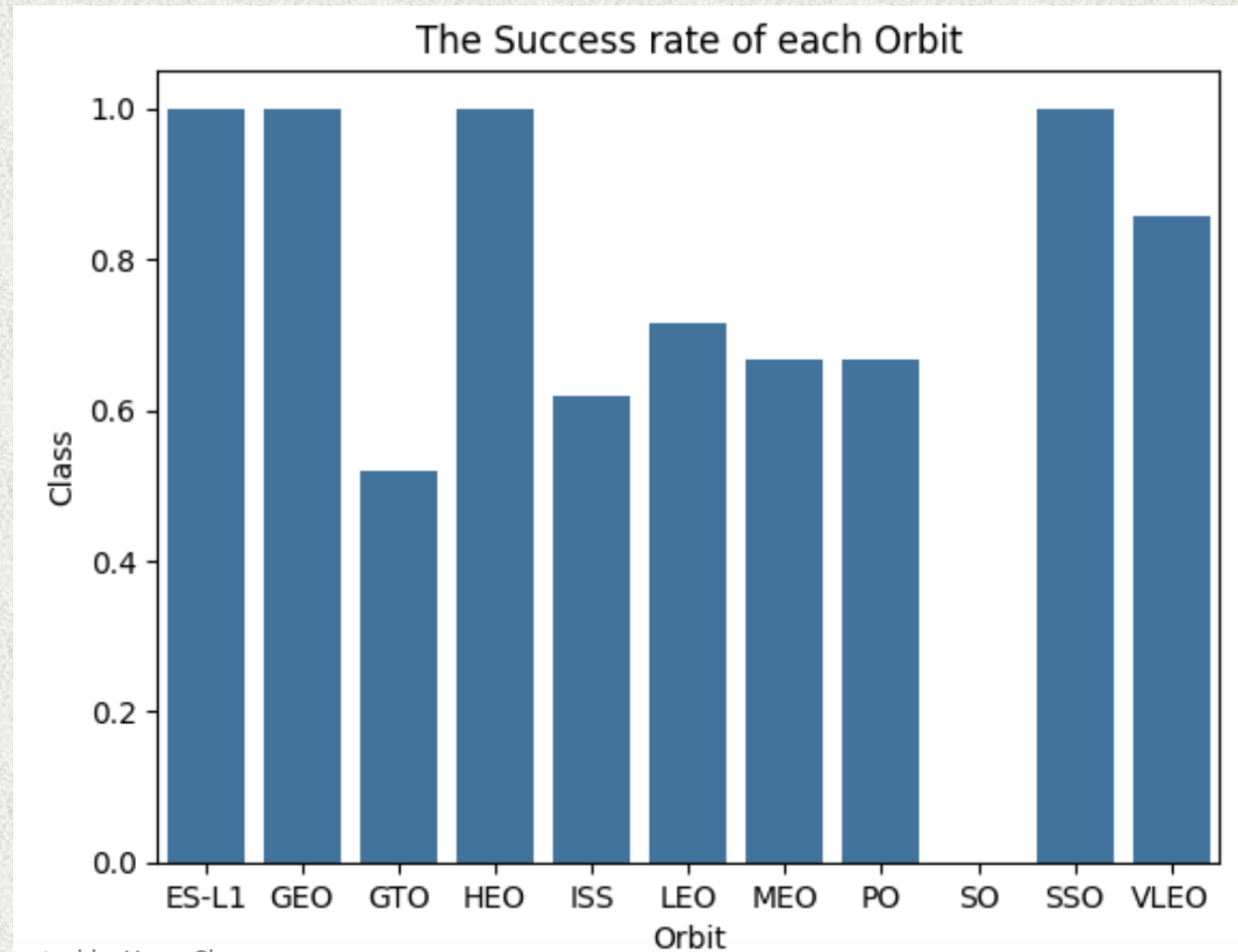
The Relationship Between Success Rate of each Orbit Type

```
sns.barplot(x="Orbit", y='Class', data=df_group2)
plt.xlabel ("Orbit")
plt.ylabel ("Class")
plt.title ("The Success rate of each Orbit")
```

Result:

The **highest** success rate:
ES-L1, GEO, HEO, SSO

The **lowest** success rate:
GTO



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The Relationship between Flight Number and Orbit types

```
sns.catplot(x="FlightNumber", y='Orbit', hue='Class', data=df)
plt.xlabel("Flight Number")
plt.ylabel("Orbit")
plt.title("The Relationship between Flight Number and Orbit types")
```

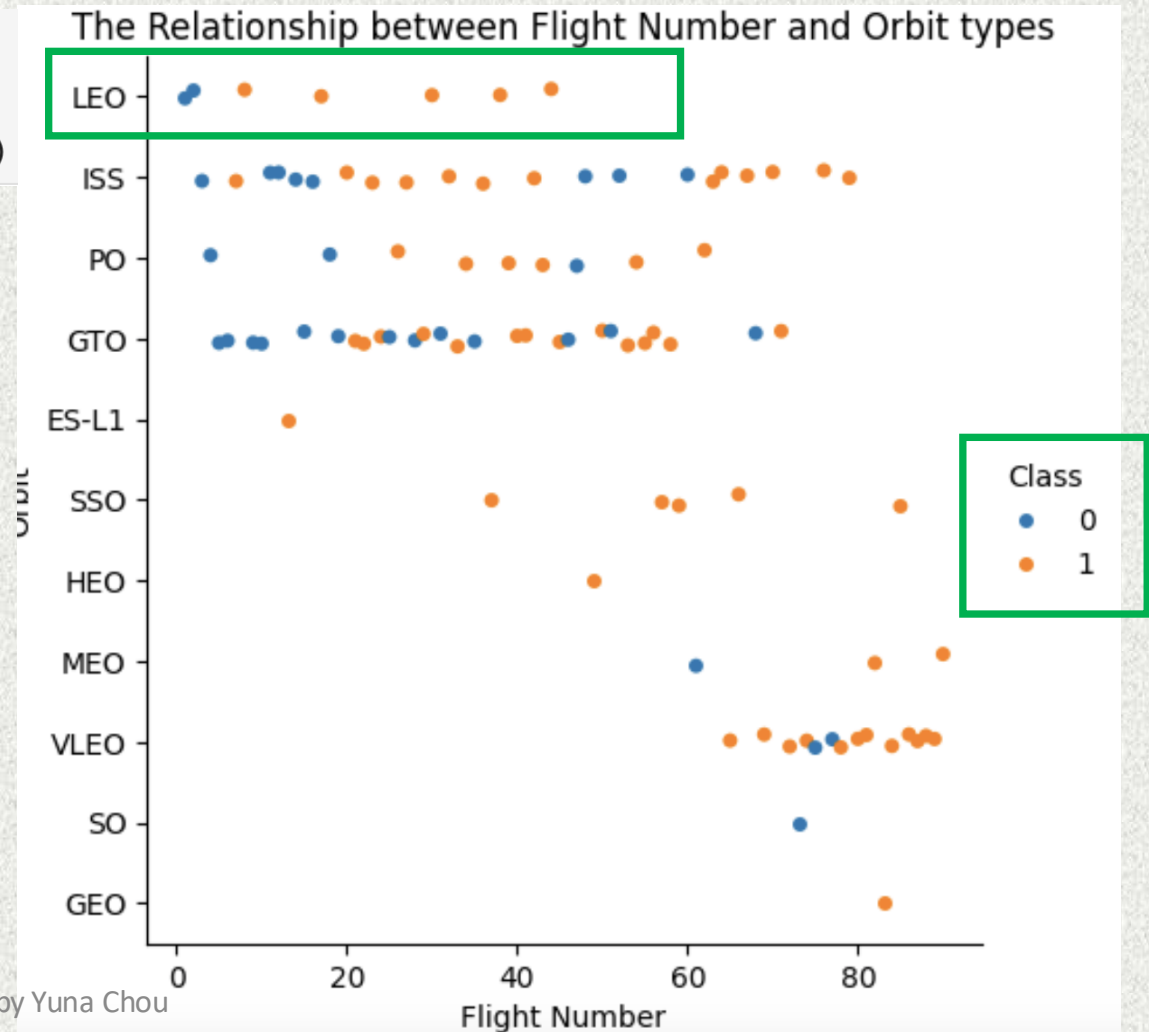
Result:

The **highest** success rate:

ES-L1, GEO, HEO, SSO

→ The flight number are less than other orbit type which they have lower success rate.

→ A **orbit type- LEO** shows that by increasing flight number, it remained success outcomes.



The Relationship between Pay load Mass (Kg) and Orbit types

```
sns.catplot(x="PayloadMass", y="Orbit", hue="Class", data=df)
plt.xlabel("Pay load Mass (Kg)")
plt.ylabel("Orbit")
plt.title("The Relationship between Pay load Mass (Kg) and Orbit types")
```

Result:

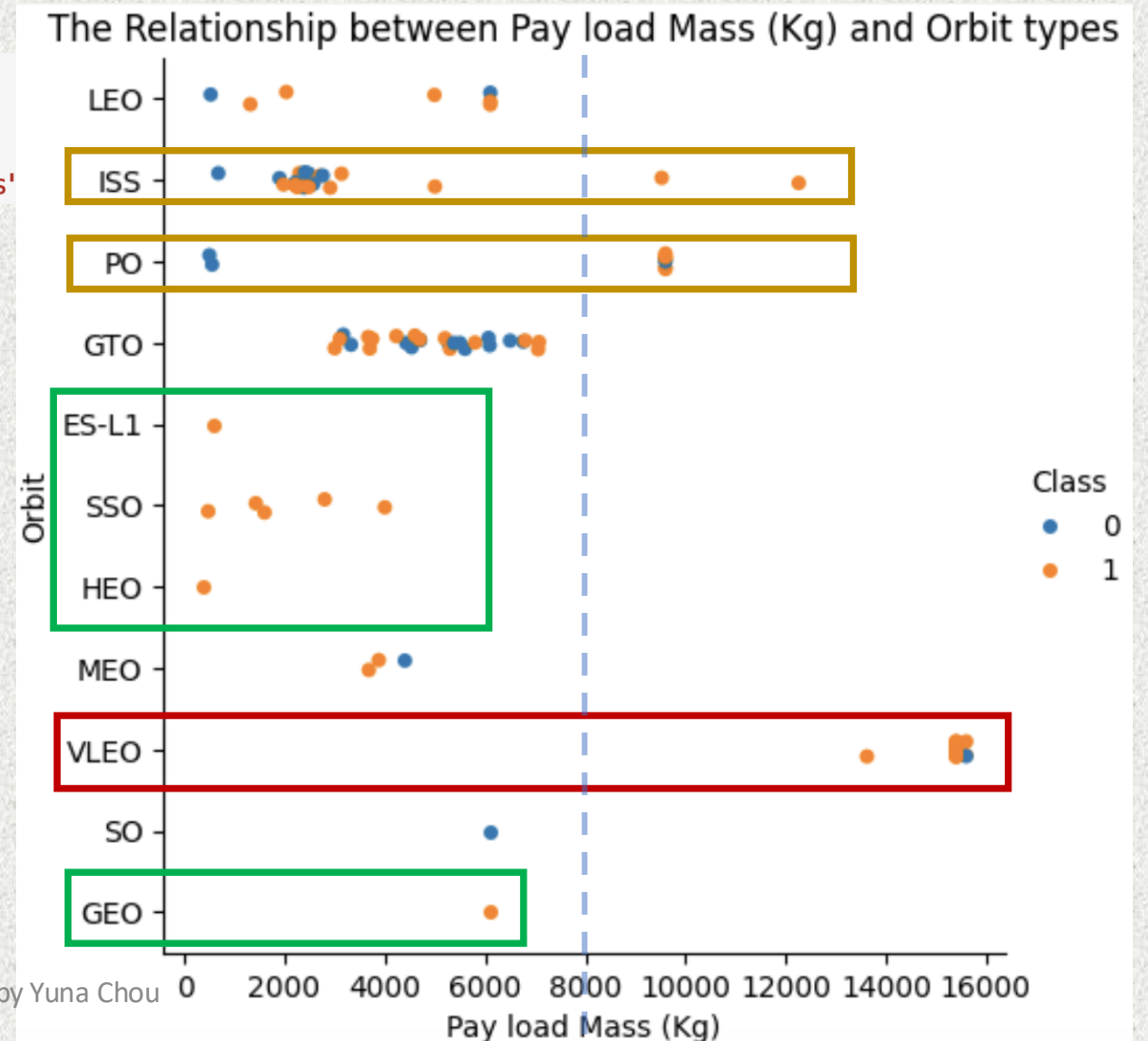
The **highest** success rate:

ES-L1, GEO, HEO, SSO

→ They all used **under 6000 kg** pay load mass, except GEO. GEO used over 6000 kg but under 7000 kg pay load mass

→ Another special observation is ISS. **ISS** data shows that it used **over 5000 kg** with successful results.

→ The data from **PO** provided two segments: under 2000 kg, the results were failed; over 9000 kg, it gained success rate.



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The Launch Success Yearly Trend

```
df_group3.plot(x='Date', y="Class", kind='line', color="blue", linewidth=2)

plt.xlabel("Year")
plt.ylabel("Class")
plt.title("The Launch Success Yearly Trend")
```

Result:

The line plot shows positive linear relationship between time (yearly) and success rate (class) from 2013 to 2020.

The success rate increased by yearly.



Selected Variables Summay

X-axis	Y-axis	3 rd parameter	Plots	Result
Flight Number	Pay Load Mass (kg)	Class	Scatter plot	Positive linear
Pay Load Mass (kg)	Launch Site	Class	Scatter plot	Observed the distribution
Orbit type	Class (Success rate)	--	Bar-plot	Observed the highest and lowest success rate orbit type
Year	Class (Success rate)	--	Line	Observed a positive trend from 2013 to 2020

Launch Site in the Map with Location Markers

```
spacex_df= spacex_df[["Launch Site", "Lat", "Long", "class"]]  
launchsites_df= spacex_df.groupby(["Launch Site"], as_index=False).first()  
launchsites_df= launchsites_df[["Launch Site","Lat","Long"]]  
launchsites_df
```

Selected the necessary variables:

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745

Launch Site on the Map

```
# Initial the map
site_map = folium.Map(location=nasa_coordinate, zoom_start=5)
# For each launch site, add a Circle object based on its coordinate (Lat, Long) values. In addition, add Launch site name as a popup label

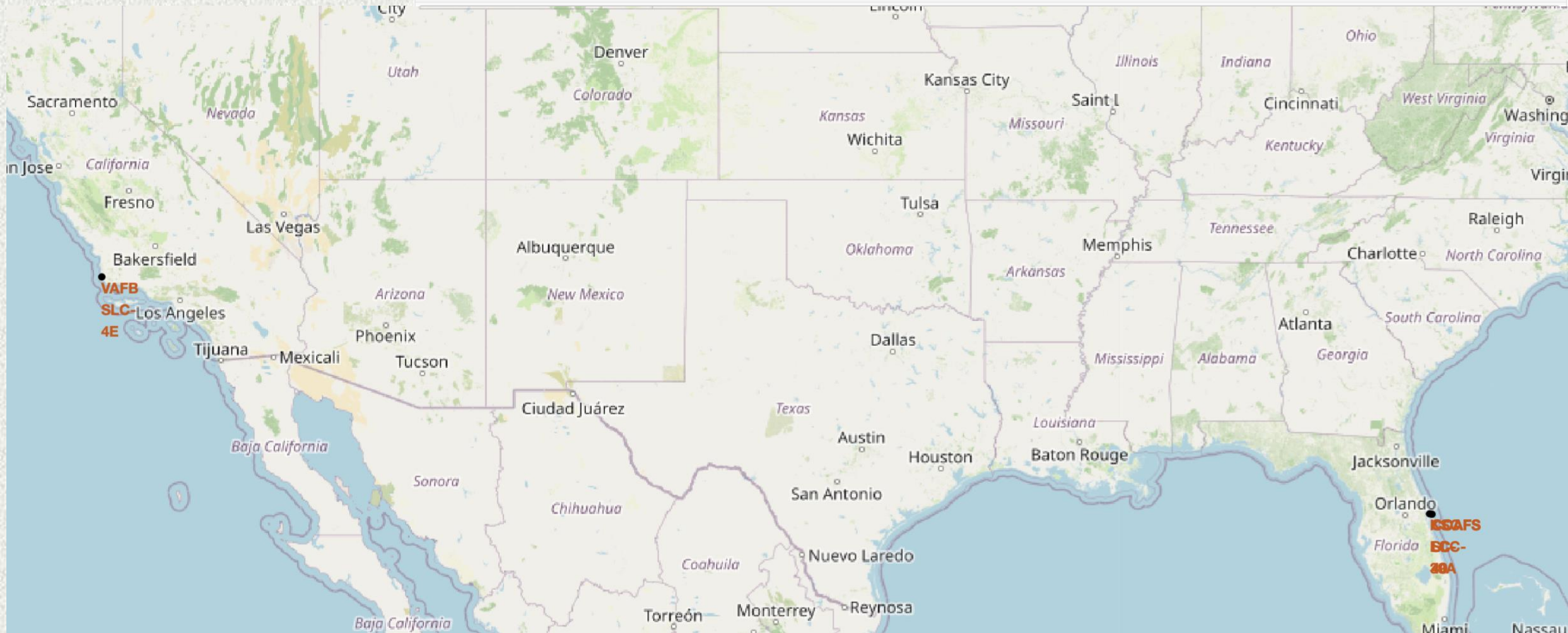
for index, row in launchesites_df.iterrows():

    # get Location
    location= [row["Lat"], row["Long"]]

    # Marker
    folium.Marker(location=location,
                  icon=DivIcon(icon_size=(20,20),icon_anchor=(0,0),
                              html='<div style="font-size: 12; color:#d35400;"><b>s</b></div>' % row["Launch Site"])).add_to(site_map)

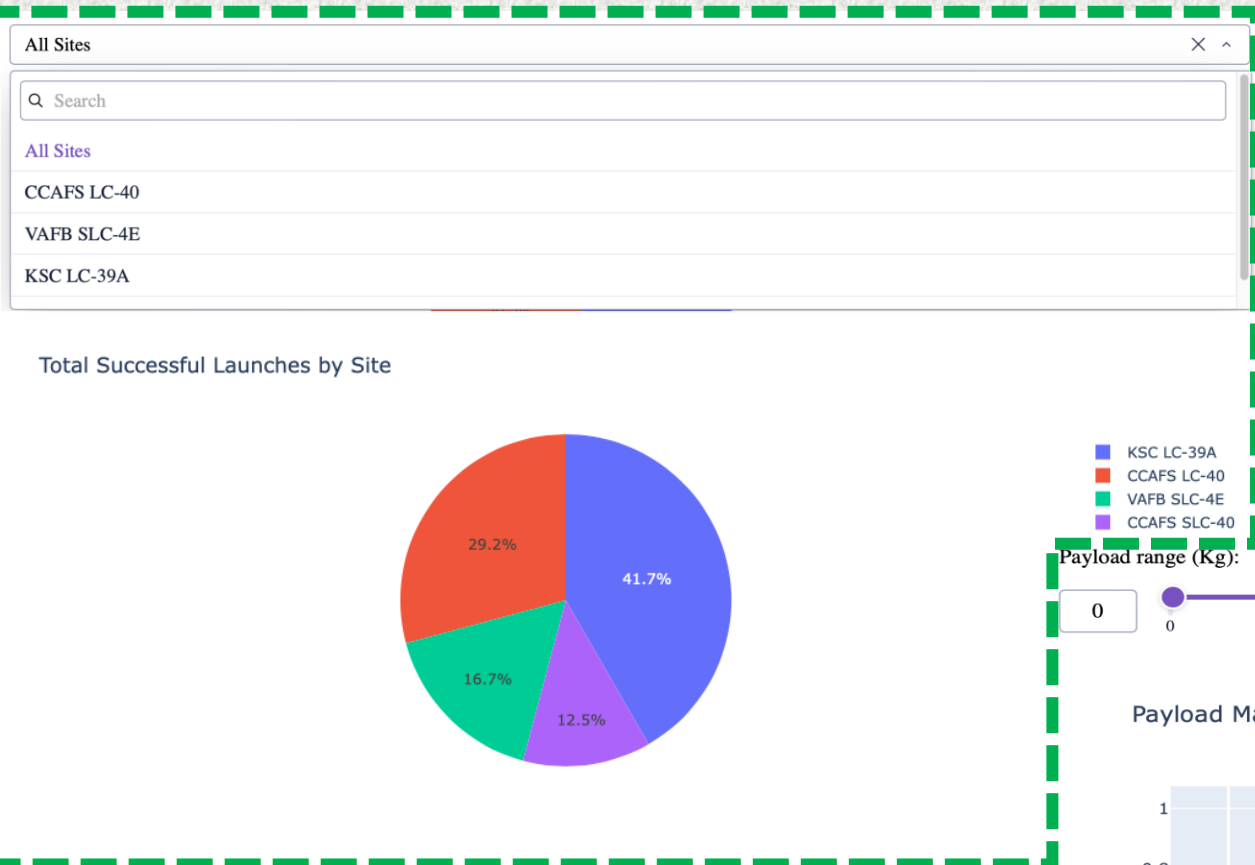
    # Circle
    folium.Circle(location= location, radius=1000, color='#000000', fill=True).add_child(folium.Popup(row["Launch Site"])).add_to(site_map)

site_map
```



Plotly Dash dashboard

- The pie chart shows the portion of successful launches by site.
- The scatter shows the distribution between payload mass and the launch success by booster version to observe whether has regular patterns.



Phase 2 – Machine Learning Prediction

Selected X and Y Attributes

1. Y- [Class]: 1 or 0

1: success landed

0: false landed

2. X- Standardize the data with Numpy array

3. Modeling selections:

✓ Logistic Regression

✓ SVM

✓ Decision Tree

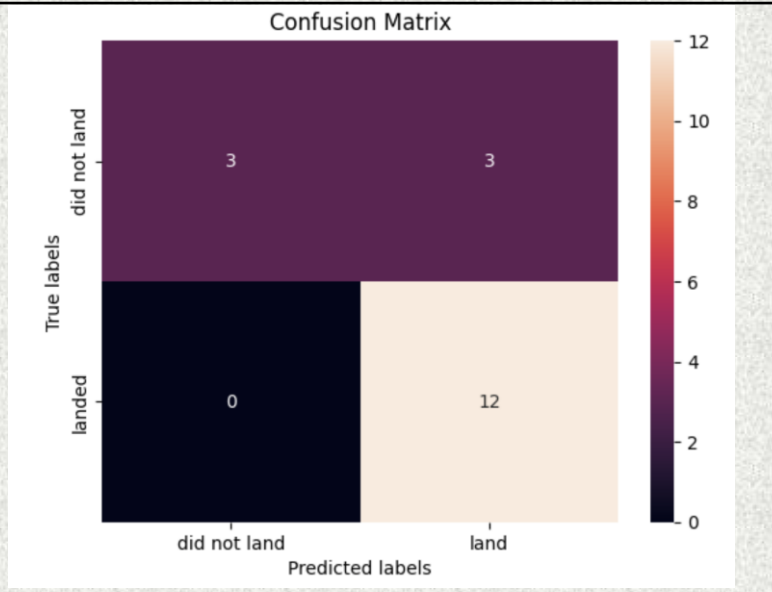
✓ KNN

Machine Learning Prediction Outcomes

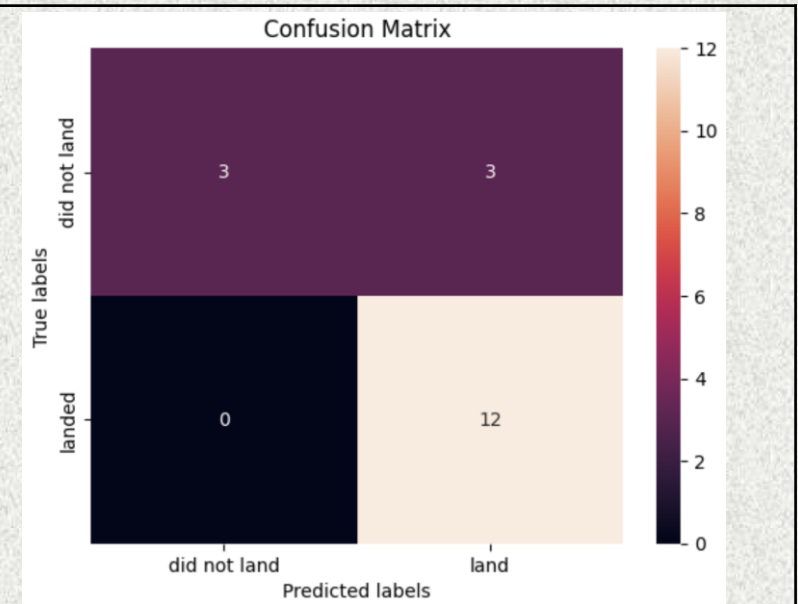
Methods (Grid Search CV)	Best Parameter		Best Score	Accuracy Score	Precision	Recall	F1_score	Confusion Matrix			
								TP	TN	FP	FN
Logistic Regression	C	0.01	0.8464	0.8333	0.8000	1	0.8889	12	3	3	0
	penalty	L2									
	solver	lbfgs									
SVM	C	1.0)	0.8482	0.8333	0.8000	1	0.8889	12	3	3	0
	gamma	0.0316									
	kernel	sigmoid									
Tree-Decision	criterion	gini	0.875	0.8333	0.8000	1	0.8889	12	3	3	0
	max_depth	4									
	max_features	sqrt									
	min_samples_leaf	2									
	min_samples_split	2									
	splitter	best									
KNN	algorithm	auto	0.8482	0.8333	0.8000	1	0.8889	12	3	3	0
	N neighbors	10									
	p	1									

Confusion Matrix

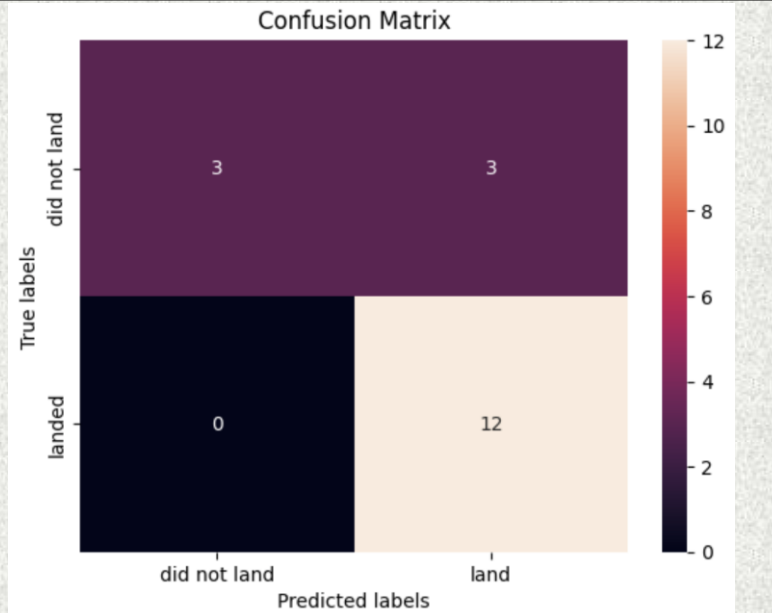
**Logistic
Regression**



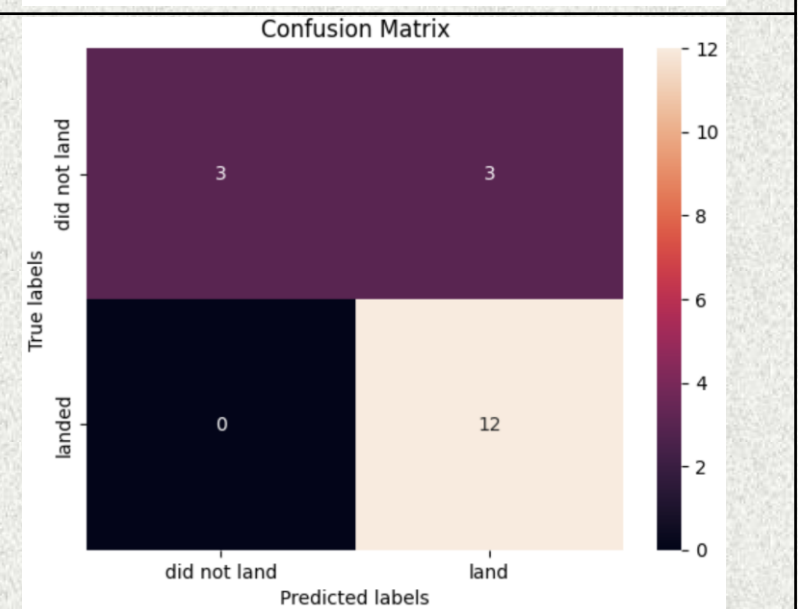
**Decision
Tree**



SVM



KNN



Conclusion

- **Decision Tree** performed best because it achieved the highest cross_validation score (0.875).
- As a result, decision tree showed the strongest cross_validation performance, but the test results indicate no practical performance difference among the four models because they generated identical predictions on the test set.
- Although Decision Tree achieved the highest cross-validation score (0.875), all four models produced identical test-set predictions, resulting in the same Accuracy (83.3%), Precision (80%), Recall (100%), and F1 Score (88.9%). This suggests that the dataset is either relatively easy to classify or the test set is too small to reveal meaningful performance differences among the models

Additional Insights

- The sample size is small, it has not been identified that we could use the models as a perfect prediction model.
- In overall, I used data to observe the successful rate depends on the variables based on the sizes, weights, locations and time with classification methods for selecting the best prediction model.
- There are many factors to affect the successful rate of safe landed including materials, electronic mechanism and chassis design, weather conditions, operating workflows, systems compatibility, and other effective potential parameters. I suggest that collect more data to validate the prediction model and defined the most correlated factors that affect the success outcomes.

Appendix

- IBM Skills Network Lab
- IBM Data Science with Python Professional Certificate
- <https://www.spacex.com/vehicles/falcon-9>